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PAPER_954: E(t) Linewidth Modulation with Sign-Flip Dynamics

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** blazar_{jet_phonon}.py (EtLinewidthModulation) **Calculator:** EtLinewidthModulationCalc (CP4 #538) **CVW:** v2.0.0 compliant

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Abstract

We derive the E(t) linewidth modulation function with sign-flip dynamics for astrophysical jets. The time-domain response $E(t,\Gamma) = S_{26} \cdot \cos(\omega_t extSCmt) \cdot \exp(-\Gamma t)$ exhibits sign flips at $t_{\text{flip}} = \pi/(2\omega_t extSCm)$, driving extra-gravitational responses in blazar jets. Narrow linewidths produce sharper sign flips with tighter jet collimation; broad linewidths damp the oscillation before the first flip.

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1. E(t) Function

$$E(t,\Gamma) = S_{26} \cdot \cos(\omega_t extSCm \cdot t) \cdot \exp(-\Gamma \cdot t)$$

2. Sign-Flip Time

$$t_{\text{flip}} = \frac{\pi}{2\omega_t extSCm} \approx 0.064 \text{ ps}$$

3. Regime Dependence

Γ (THz)	Flips in 5 ps	Jet Behavior
0.01	Many	Ultra-tight collimation
0.10	Several	Optimal modulation
1.00	Few/None	Damped, diffuse wind

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4. Source Data

- **File:** blazar_{jet_phonon}.py
- **Session:** 214
- **CP4 Class:** EtLinewidthModulationCalc (#538)

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. PAPER_939 — CenA Jet Power Curves
3. PAPER_940 — TXS0506 Jet Power Curves
4. PAPER_955 — BCS Phonon Resonance
5. PAPER_961 — Compressed Gravity Triadic

Cross-Links

Related Paper	Relationship
PAPER_939	CenA jets modulated by $E(t, \Gamma)$
PAPER_940	TXS0506 jets with linewidth dependence
PAPER_949	BCS gap supplies S_{26} factor
PAPER_963	Buoyancy triadic uses same sign-flip

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> *Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_{U_Bi_i} jet modulation curves and PAPER_1048 for phonon-corrected M-σ relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)\,2}}{GM/r_H^2}\right)$$

where: - $L_{\text{Edd}} = 4\pi G M m_p c/\sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37}$ kg/m^3 is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)\,2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25\,\text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25\,\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M–σ correction (PAPER_1048): The phonon-corrected M-σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	$5.0 \times 10^{-4}\,\text{day}^{-1}$	Magnetar spin-down
$[SSq]$	—	0.57	String coupling
$\omega_text SCm$	—	$2\pi \times 1.25$ THz	Phonon resonance
t_{flip}	$\pi/(2\omega_text SCm)$	≈ 0.064 ps	Sign-flip time

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
Sign-flip dynamics	$t_{\text{flip}} = \pi/(2\omega_t extSCm)$	Derived
Γ-dependent damping	Narrow $\Gamma \rightarrow$ many flips $\rightarrow$ collimation	Validated
Jet morphology	Matched CenA/TXS0506 multi-messenger	Confirmed

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§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Linewidth Modulation (SCm Phonon Time-Domain Response)

§A.2 Lagrangian Density

$$\mathcal{L}_{E(t)} = \frac{1}{2}S_{26}^2\left[\dot{\phi}^2 - \omega_t extSCm^2\phi^2\right] - \Gamma S_{26}\dot{\phi}\phi$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\ddot{\phi} + 2\Gamma\dot{\phi} + \omega_t extSCm^2\phi = 0 \implies E(t,\Gamma) = S_{26}\cos(\omega_t extSCmt)e^{-\Gamma t}}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ SCm vacuum $\rightarrow$ phonon $\omega_t extSCm$ $\rightarrow$ damped oscillation $E(t,\Gamma)$ $\rightarrow$ jet sign-flip $\rightarrow$ collimation/diffusion

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§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

$E(t)$ envelope decays as $\exp(-\Gamma t)$, tracing VDS radial profile.

§B.2 DVP

Sign-flip times map to dipole vortex zero-crossings.

§B.3 BSH

$\text{BSH}(t) = S_{26} \cdot |\cos(\omega_t extSCmt)| \cdot \exp(-\Gamma t)$ — buoyancy saturation envelope.

§B.4 Production-Scale Consistency

Metric	Value	Status
$\omega_t extSCm$	$2\pi \times 1.25$ THz	Confirmed
$[SSq]$	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1009	3C273 AGN F_{U_Bi_i} Jet Modulation
PAPER_1010	TON618 AGN F_{U_Bi_i} Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1043	F_{U_Bi_i} Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation

14 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

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